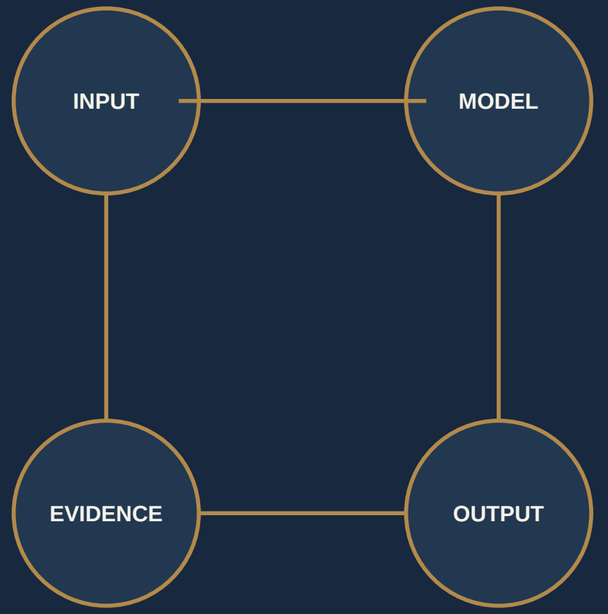
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Evolutionary Robotics

AI.401 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define genetic algorithms.
- Apply key methods including evolutionary optimization.
- Build or analyze a practical system involving genetic programming.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Genetic algorithms

02

METHODS

Evolutionary optimization

03

SYSTEMS

Genetic programming

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Genetic algorithms

Genetic algorithms

02 · Evolutionary optimization

Evolutionary optimization

03 · Genetic programming

Genetic programming

04 · Evolutionary morphology and ro

Evolutionary morphology and robot behavior evolution

05 · Mandatory practical: Evolve ro

Mandatory practical: Evolve robot controllers or robot designs through simulation

Core ideas, made explicit

AI.401 • Distinguish definitions, assumptions and evidence

- Genetic algorithms
- Evolutionary optimization
- Genetic programming
- Evolutionary morphology and robot behavior evolution
- Mandatory practical: Evolve robot controllers or robot designs through simulation

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.401 • Distinguish definitions, assumptions and evidence

- Genetic algorithms
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- Mandatory practical: Evolve robot controllers or robot designs through simulation

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

Genetic algorithms

INPUTS

Evolutionary optimization

METHOD

Genetic programming

OUTPUT

Evolutionary morphology and robot behavior evolution

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving genetic algorithms.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using genetic algorithms.
- Implement or simulate a baseline using evolutionary optimization.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.